SP20

DEVELOPMENT, EXPERIENCE & MOTOR RECOVERY LAB

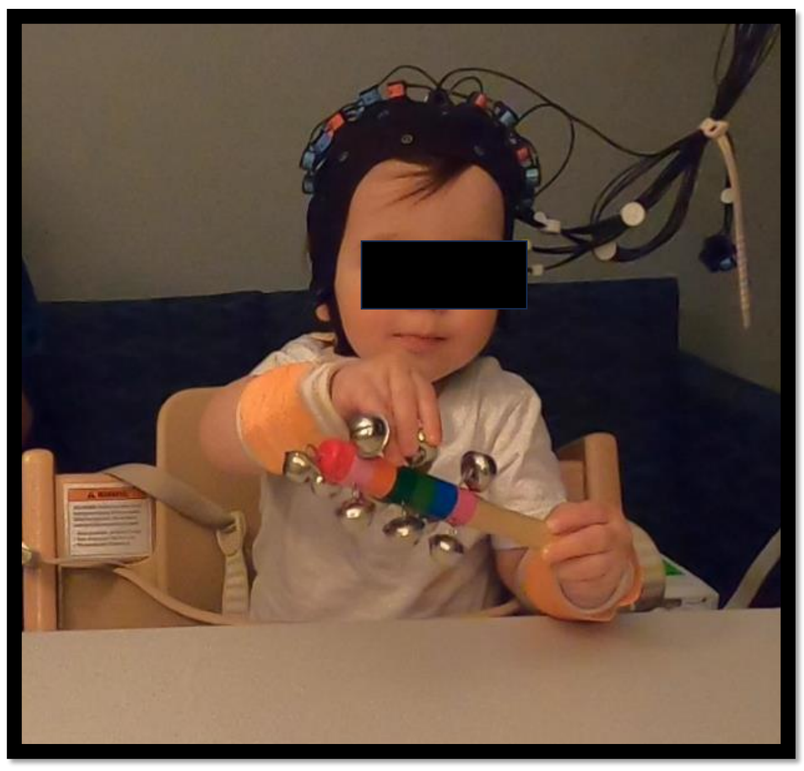
Longitudinal Trajectories of Bilateral Hand Use: Changes in Behavior and Cortical Activity

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BACKGROUND: Object manipulation skills in infants cascade across developmental domains. Infants first demonstrate symmetric bimanual exploration (4-5mo), and then later (11 mos. and beyond) engage in role-differentiated bimanual manipulation (RDBM). RDBM is a skill in which each hand performs a separate, but complementary role, during bimanual manipulation. RDBM is essential for performance of functional skills (e.g. buttoning a shirt). The development of RDBM is thought to reflect a shift in the organization of motor control and underlying neurophysiology of the two hemispheres.

OBJECTIVE: To investigate trajectories of cortical activity during the development of RDBM.



METHODS: Functional near-infrared spectroscopy (fNIRS) was used to obtain measures of functional brain activity during semi-structured play sessions in a longitudinal sample (n=26) of infants from 6-24 months. Sessions were coded for bouts of contact and RDBM.

Figure 1. Manual exploration becomes increasingly bilateral

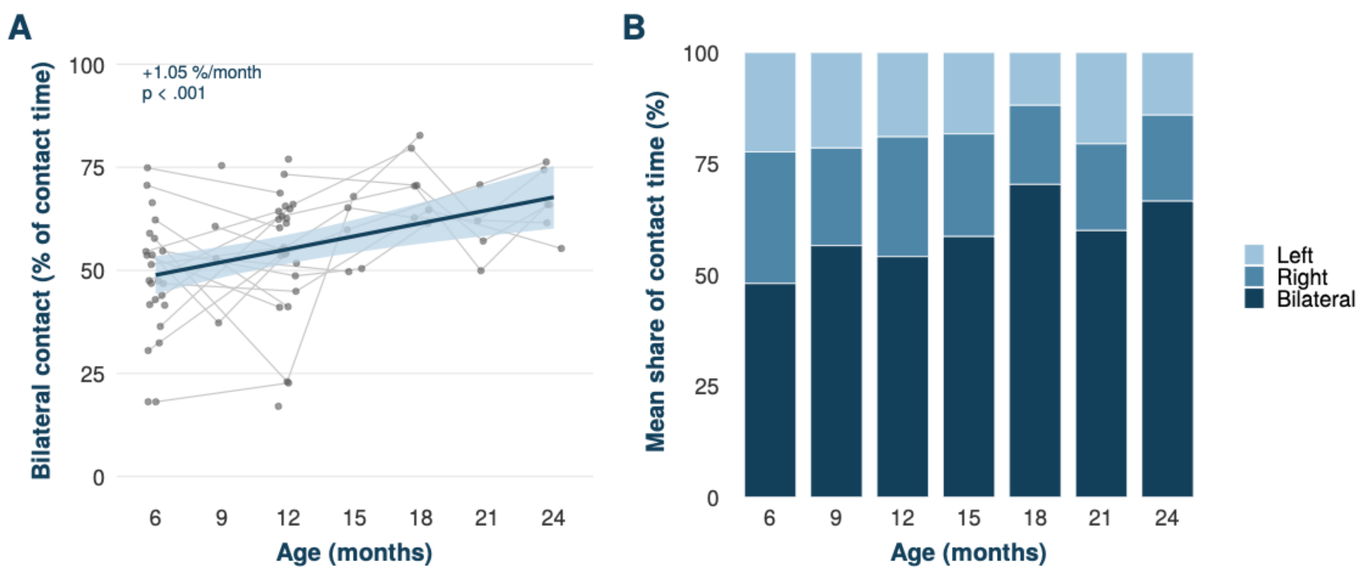


Figure 2. Bout-level decomposition of RDBM

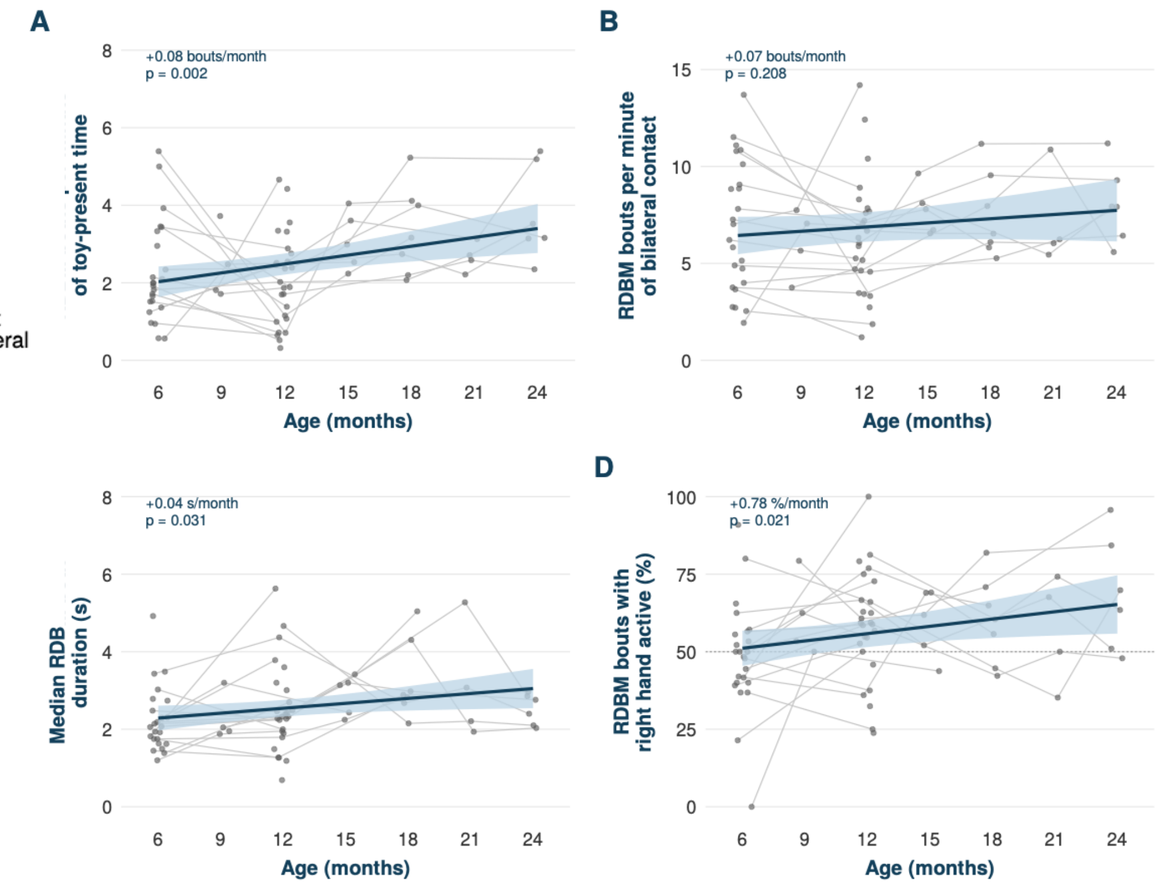


Figure 3. Cortical lateralization of the manipulating hand during RDBM

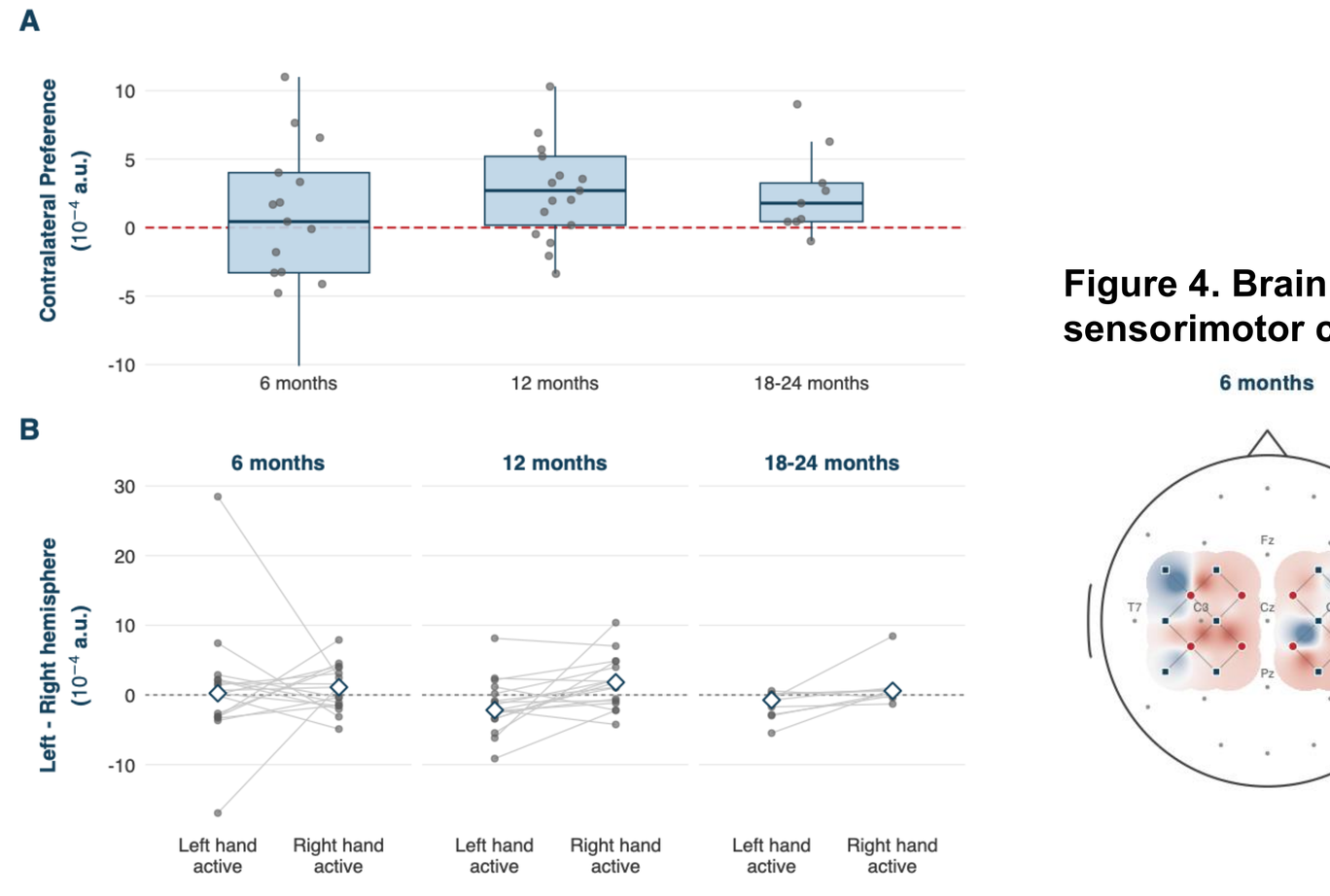
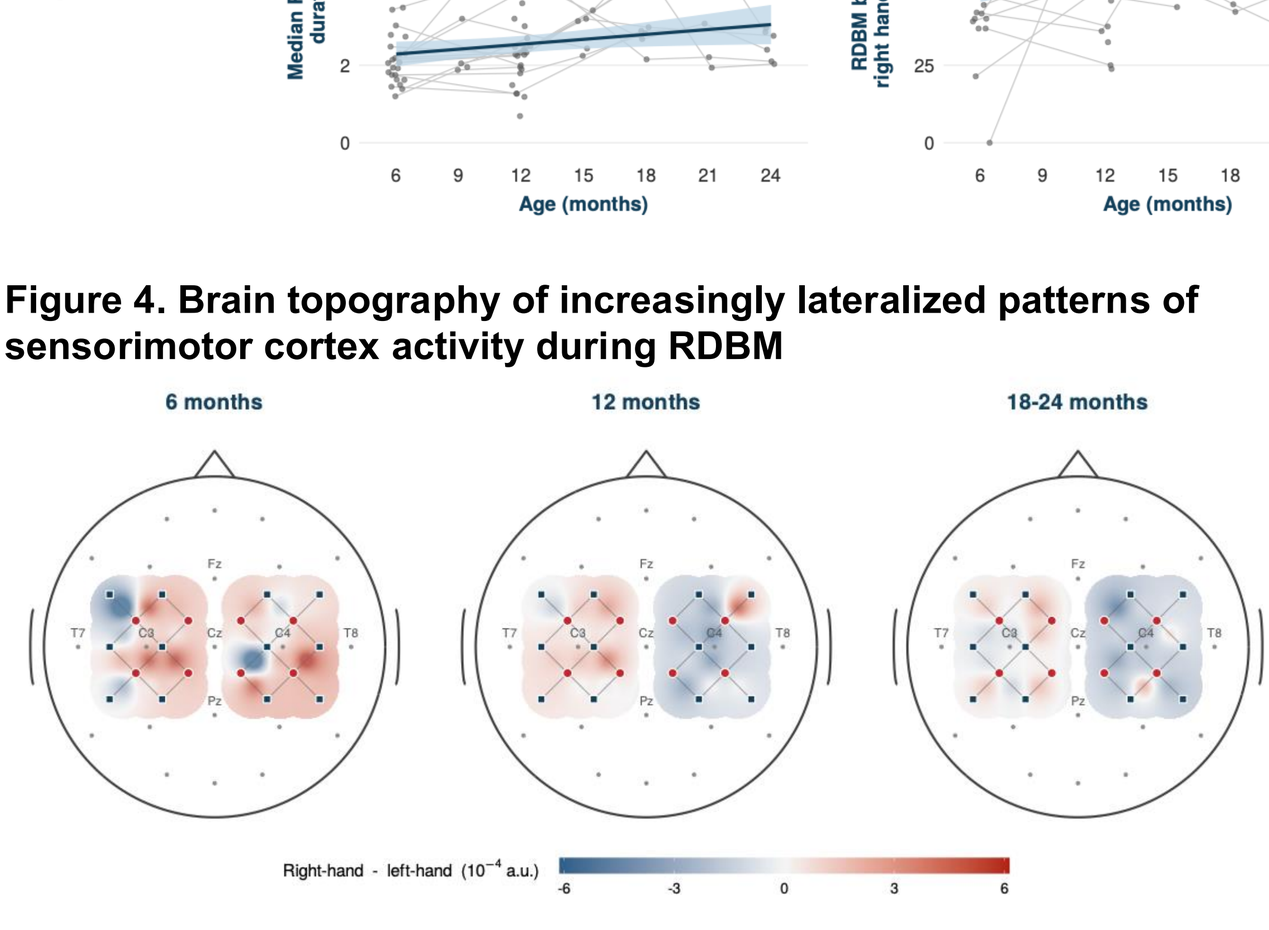


Figure 4. Brain topography of increasingly lateralized patterns of sensorimotor cortex activity during RDBM



CONCLUSIONS: Infants produce more RDBM at older ages because they increase the amount of bimanual manipulations with age. fNIRS data show greater contralateral cortical activation at 12 months and older compared to 6 months. This demonstrates lateralization of motor control of the active hand in RDBM.

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